

BRESCIA MONTICHIARI, ItalyWMO: **162593**Lat: **45.429N**Lon: **10.331E**Elev: **356**StdP: **14.51**Time Zone: **1.00 (EUC)**Period: **04-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.5	26.7	11.8	10.2	39.0	16.8	13.0	37.1	20.0	48.1	16.9	43.6	4.7	280	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	91.8	71.2	89.6	70.5	87.6	69.4	73.6	87.1	72.4	85.5	71.1	83.6	6.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.6	110.2	80.8	68.0	104.3	78.9	66.5	98.7	77.9	37.3	87.0	36.2	85.7	35.2	84.4	79.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	15.4	13.0	19.4	95.8	4.1	2.3	16.5	97.4	14.1	98.8	11.8	100.1	8.9	101.8		
(5)			19.0	75.8	4.0	2.1	16.1	77.3	13.8	78.5	11.5	79.7	8.6	81.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.8	37.7	41.1	49.5	57.8	64.9	73.3	77.5	76.0	68.5	58.8	48.2	38.7	
	DBStd	14.96	5.03	5.43	5.70	5.36	5.92	5.55	4.35	5.07	5.15	5.51	6.03	5.60	
	HDD50	1170	383	251	77	5	0	0	0	0	0	4	100	350	
	HDD65	3841	847	670	479	221	76	5	0	1	23	201	503	815	
	CDD50	4005	1	1	62	240	463	700	852	805	555	278	47	1	
	CDD65	1198	0	0	0	6	73	255	387	340	128	9	0	0	
	CDH74	10873	0	0	4	62	553	2303	3919	3230	770	32	0	0	
	CDH80	3874	0	0	0	4	119	807	1543	1231	167	3	0	0	
Wind	WSAvg	5.8	5.1	5.6	6.5	6.6	6.5	6.2	6.3	5.9	5.7	5.1	5.6	4.9	
Precipitation	PrecAvg	35.3	2.6	2.3	2.6	2.6	3.3	3.2	2.9	3.4	2.7	3.6	3.7	2.2	
	PrecMax	49.8	6.1	5.0	6.3	4.6	5.9	7.1	7.0	7.2	8.2	11.7	6.1	3.7	
	PrecMin	24.3	0.2	0.0	0.0	1.1	0.4	0.4	0.6	0.4	0.2	0.0	1.1	0.8	
	PrecStd	7.1	1.8	1.4	1.8	1.1	1.9	1.9	1.9	2.0	2.4	3.1	1.6	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.3	60.9	72.0	79.1	87.6	93.2	95.0	95.2	87.7	78.4	64.8	55.4	
		MCWB	49.5	49.1	54.2	60.5	65.7	70.7	71.7	71.4	67.8	63.6	58.5	50.2	
	2%	DB	50.3	55.4	67.6	75.3	82.8	89.9	91.7	92.8	84.5	73.5	61.1	51.9	
		MCWB	45.0	47.1	53.0	58.9	64.2	70.2	71.7	70.8	67.5	62.4	56.3	47.8	
	5%	DB	48.0	52.1	64.0	71.9	80.2	87.6	89.6	89.3	82.0	71.2	58.9	49.7	
		MCWB	43.6	45.3	51.8	57.4	63.3	69.4	70.9	70.2	66.6	61.5	55.0	46.7	
	10%	DB	45.0	49.7	60.5	69.5	76.9	84.5	87.6	86.2	78.7	68.0	56.9	46.8	
		MCWB	42.4	44.4	50.2	56.7	62.0	67.6	69.9	69.1	64.2	60.0	53.5	44.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.5	50.6	56.4	62.3	68.5	74.5	75.6	75.0	71.0	66.3	60.5	51.9
			MCDB	54.7	57.7	65.5	74.1	81.1	88.7	89.0	88.0	82.0	72.4	63.0	53.1
2%		WB	46.8	48.5	54.7	60.5	66.4	72.6	73.8	73.1	69.5	64.6	57.6	49.2	
		MCDB	48.6	53.1	64.1	71.8	78.8	86.4	87.4	86.0	80.3	70.5	60.3	50.3	
5%		WB	44.8	46.9	53.0	59.1	64.8	70.7	72.6	72.0	68.0	63.2	55.8	46.8	
		MCDB	46.2	50.3	61.1	69.4	76.4	83.3	85.9	84.9	77.8	68.6	58.3	48.5	
10%		WB	42.9	45.4	51.4	57.5	63.2	69.2	71.3	70.6	66.6	61.5	54.3	44.9	
		MCDB	44.4	48.1	58.0	67.2	74.3	81.7	83.9	83.0	76.2	66.4	56.6	46.2	
Mean Daily Temperature Range		5% DB	MDBR	11.7	13.8	16.5	16.5	17.0	17.2	17.6	17.3	16.0	13.7	11.1	11.7
			MCDBR	16.4	19.3	23.1	21.7	20.2	19.6	19.0	19.5	18.9	17.3	13.5	14.9
	MCWBR		11.5	12.2	11.9	9.5	7.5	7.0	6.7	6.2	6.3	7.9	8.8	10.7	
	5% WB	MCDBR	12.8	15.7	19.1	19.6	18.5	18.4	17.2	17.8	16.5	13.9	11.1	11.5	
		MCWBR	9.9	10.5	10.6	9.2	7.9	7.5	6.8	6.8	6.3	7.1	7.5	8.9	
Clear-Sky Solar Irradiance	taub		0.348	0.378	0.425	0.460	0.458	0.484	0.466	0.454	0.444	0.432	0.378	0.342	
	taud		2.375	2.260	2.135	2.072	2.128	2.103	2.163	2.208	2.201	2.186	2.302	2.401	
	Ebn at Noon		236	249	252	253	259	252	255	253	243	225	219	225	
	Edh at Noon		24	33	42	49	48	49	46	42	39	35	26	22	
All-Sky Solar Radiation	RadAvg		451	706	1122	1428	1766	2021	2071	1766	1307	788	466	385	
	RadStd		70	105	150	158	162	110	114	115	87	114	70	51	